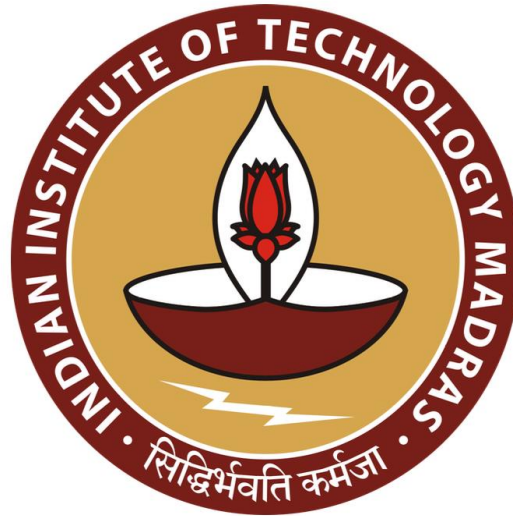


Project Report
for the Subject
Market Research - Jan 2026



**Topic - Survey-Based Analysis on
Customer Satisfaction Data in Loan Services**

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1. EXPLORATORY DATA ANALYSIS (EDA)

1.1 Dataset Overview

Before answering the case questions, the dataset was examined for data quality, distributions and intervariable relationships. The dataset consists of 200 respondents with no missing values.

Attribute	Detail
Total Respondents	200
Zones	South (65), East (53), North (47), West (35)
Missing Values	0 (complete dataset)
Score Range	0 to 9 (scale of 0-10; max observed = 9)
Parameters	9 service parameters + satisfaction + recommend + comeback

1.2 Descriptive Statistics

Variable	Mean	Std	Min	25%	Median	75%	Max	Skew
Satisfaction	7.91	1.80	0	8.0	8.0	9.0	9	-2.75
Recommend	7.84	2.25	0	8.0	8.5	9.0	9	-2.82
Comeback	7.85	2.26	0	8.0	9.0	9.0	9	-2.80
Overall Experience	8.41	1.55	0	9.0	9.0	9.0	9	-3.74
Loan & Charges	8.37	1.72	0	9.0	9.0	9.0	9	-3.76
Sanction Terms	8.37	1.72	0	9.0	9.0	9.0	9	-3.76
Documentation	8.38	1.72	0	9.0	9.0	9.0	9	-3.79
Time - Loan	8.41	1.67	0	9.0	9.0	9.0	9	-3.97
Time - Welcome Letter	7.76	2.69	0	8.0	9.0	9.0	9	-2.28
Staff Politeness	8.44	1.61	0	9.0	9.0	9.0	9	-3.86
Staff Availability	8.44	1.61	0	9.0	9.0	9.0	9	-3.86
Staff Knowledge	8.44	1.61	0	9.0	9.0	9.0	9	-3.86

Key EDA Findings:

- All variables show strong negative skew - scores cluster near 8-9, a ceiling effect common in financial services satisfaction surveys.
- Staff parameters (politeness, availability, knowledge) are almost perfectly correlated ($r \approx 1.0$) - respondents perceive them as a single construct.
- Loan process parameters (charges, sanction terms, documentation) also cluster tightly ($r > 0.95$).

- Time to receive a welcome letter is the most independent and most variable parameter (Std = 2.69).

2. Q1: MEAN SCORE OF SATISFACTION

2a. Overall Mean Satisfaction

Metric	Value	Interpretation
Mean Satisfaction	7.91 / 10	Good (benchmark: 8.0+ = strong in financial services)
Standard Deviation	1.80	Moderate spread around mean
95% Confidence Interval	[7.66, 8.16]	True population mean likely in this range
Median	8.0	50% of respondents scored 8 or above
Mode	9	Most common score indicates positive signal
Top-2-Box (scores 8–9)	81.0%	Strong: 4 in 5 customers highly satisfied

2b. Zone-wise Mean Satisfaction

Zone	n	Mean	Std Dev	Median	Rank
South	65	8.23	1.67	9.0	#1 ~ Best
East	53	7.98	1.55	8.0	#2
West	35	7.63	1.85	8.0	#3
North	47	7.60	2.14	8.0	#4 ~ Needs Attention

3. Q2: NET PROMOTER SCORE (NPS)

NPS is computed from the "Willingness to Recommend" question.
Scores of 9-10 = Promoters; 7-8 = Passives; 0-6 = Detractors.

$$NPS = \% \text{ Promoters} - \% \text{ Detractors.}$$

3a. Overall NPS

Category	Count	Percentage
Promoters (Score 9-10)	100	50.0%
Passives (Score 7-8)	79	39.5%
Detractors (Score 0-6)	21	10.5%
Overall NPS	-	+39.5 (GREAT)

3b. Zone-wise NPS

Zone	n	Promoters %	Passives %	Detractors %	NPS	Status
South	65	67.7%	26.2%	6.2%	+61.5	Great
West	35	37.1%	57.1%	5.7%	+31.4	Great
East	53	41.5%	47.2%	11.3%	+30.2	Great
North	47	44.7%	36.2%	19.1%	+25.5	Good (at risk)

- South zone is exceptional at +61.5 - driven by 67.7% promoter share.
- North zone is the weakest (+25.5) with 19.1% detractors - nearly double the average.
- West has the highest passive share (57.1%) - the largest conversion opportunity per zone.

4. Q3: PARAMETER-LEVEL MEAN SCORES

Nine service parameters from the questionnaire were analysed. Parameters with mean ≥ 8.3 are flagged as strong; below 8.0 as needing attention.

4a. Overall Parameter Means (Ranked)

Rank	Parameter	Mean	Std Dev	Status
1	Staff Politeness	8.44	1.61	Strong
1	Staff Availability	8.44	1.61	Strong
1	Staff Knowledge & Ability	8.44	1.61	Strong
4	Overall Loan Experience	8.41	1.55	Strong
4	Time to Receive Loan	8.41	1.67	Strong
6	Documentation Help	8.38	1.72	Strong
7	Loan & Charges Explanation	8.37	1.72	Strong
7	Sanction Terms Explanation	8.37	1.72	Strong
9	Time to Receive Welcome Letter	7.76	2.69	Needs Improvement

4b. Zone-wise Parameter Means

Parameter	East	North	South	West
Overall Loan Experience	8.36	8.34	8.46	8.49
Loan & Charges Explanation	8.21	8.40	8.35	8.60
Sanction Terms Explanation	8.21	8.40	8.35	8.60
Documentation Help	8.25	8.40	8.35	8.60
Time to Receive Loan	8.23	8.64	8.38	8.43

Time - Welcome Letter	7.64	8.26	7.55	7.66
Staff Politeness	8.42	8.55	8.37	8.46
Staff Availability	8.42	8.55	8.37	8.46
Staff Knowledge	8.42	8.55	8.37	8.46

- Welcome Letter delivery scores below 8.0 in East (7.64), South (7.55), and West (7.66) indicate a systemic issue.
- North scores 8.26 on welcome letter but has the worst overall satisfaction indicates a broader service quality issue.
- Staff dimensions score above 8.3 in all zones indicate a consistent strength.

5. Q4: EXECUTIVE SUMMARY

The following summarizes our findings The following observations are from the analysis of 200 loan customers surveyed across four zones.

Headline Metrics

Metric	Value
Mean Satisfaction	7.91 / 10
Net Promoter Score	+39.5 (Great)
Top-2-Box	81.0%
Detractor Rate	10.5%

Strengths

- Staff dimensions rank #1 at 8.44/10 - polite, available and knowledgeable
- South zone delivers near world-class NPS (+61.5)
- 81% of customers in Top-2-Box (scored 8 or 9)
- Loan processing speed meets expectations (8.41)

Concerns

- Welcome letter delivery is the only sub-8.0 parameter (7.76) - a hygiene failure
- North zone detractor rate (19.1%) is 2x the overall average
- 39.5% passive segment is vulnerable to competitive churn
- All parameters outscore overall satisfaction - a "global evaluation discount" is at play

6. Q5: T-TEST ~ SATISFACTION VS EACH PARAMETER

Method: Paired-sample t-test. H0: mean satisfaction = mean parameter. H1: means differ (two-tailed, $\alpha = 0.05$). Cohen's d: small < 0.2, medium 0.2–0.8, large > 0.8.

Parameter	Param Mean	Difference	t-stat	p-value	Cohen's d	Significant?
Overall Loan Experience	8.41	-0.50	-5.378	< 0.001	-0.380	Yes
Loan & Charges Explanation	8.37	-0.46	-4.003	< 0.001	-0.283	Yes
Sanction Terms Explanation	8.37	-0.46	-3.936	< 0.001	-0.278	Yes
Documentation Help	8.38	-0.47	-3.999	< 0.001	-0.283	Yes
Time to Receive Loan	8.41	-0.50	-3.776	< 0.001	-0.267	Yes
Time - Welcome Letter	7.76	+0.15	+0.722	0.471	+0.051	No
Staff Politeness	8.44	-0.53	-4.576	< 0.001	-0.324	Yes
Staff Availability	8.44	-0.53	-4.576	< 0.001	-0.324	Yes
Staff Knowledge & Ability	8.44	-0.53	-4.576	< 0.001	-0.324	Yes

Key Findings:

- 8 out of 9 parameters score significantly higher than overall satisfaction (all $p < 0.001$). This reflects a "global evaluation discount" - customers rate individual touchpoints well but discount holistically.
- Time to Receive Welcome Letter is the only exception ($p = 0.471$, not significant) - its low mean matches overall satisfaction, making it the anchor dragging the score down.
- Effect sizes are small-to-medium ($|d|$ range: 0.27–0.38).

7. Q6: CORRELATION~ SATISFACTION VS PARAMETERS

Parameter	Pearson r	R ²	Spearman ρ	p-value	Strength
Overall Loan Experience	+0.702	0.492	+0.471	< 0.001	Strong
Loan & Charges Explanation	+0.574	0.329	+0.540	< 0.001	Moderate-Strong
Sanction Terms Explanation	+0.559	0.312	+0.519	< 0.001	Moderate-Strong

Documentation Help	+0.554	0.306	+0.504	< 0.001	Moderate-Strong
Staff Politeness	+0.542	0.294	+0.482	< 0.001	Moderate
Staff Availability	+0.542	0.294	+0.482	< 0.001	Moderate
Staff Knowledge & Ability	+0.542	0.294	+0.482	< 0.001	Moderate
Time to Receive Loan	+0.420	0.176	+0.489	< 0.001	Moderate
Time - Welcome Letter	+0.193	0.037	+0.381	0.006	Weak

To identify which parameter has the greatest influence on overall satisfaction, an **Ordinary Least Squares (OLS) regression** was run with Overall Loan Experience as the independent variable and Overall Satisfaction as the dependent variable.

Parameter	Intercept	Coefficient	t-stat	p-value	R ²
Overall Loan Experience	1.076	0.813	13.857	< 0.001	0.492

- Overall Loan Experience is the strongest driver of satisfaction ($r = 0.70$, $R^2 = 0.49$).
- A 1-point improvement in loan experience is associated with a 0.813-point increase in overall satisfaction.
- This single variable explains 49.2% of the variance in satisfaction - a very high single-predictor R^2 .
- Time to Receive Welcome Letter has the weakest correlation ($r = 0.19$) - addressing it is a hygiene fix, not a delight driver.

8. Q7: STRATEGIC RECOMMENDATIONS

HIGH PRIORITY

1. Optimize the End-to-End Loan Processing Experience

Overall Loan Experience is the strongest satisfaction driver followed by Loan & Charges Explanation. A 1-point improvement here translates to a ~0.81-point increase in overall satisfaction. This parameter is the single most valuable lever the service can pull.

- Conduct process mapping to identify bottlenecks in loan disbursement workflow.
- Introduce a concierge pathway for complex or high-value loan cases.
- Set target: increase mean from 8.41 to 8.80 within two quarters.

2. Fix Welcome Letter Delivery (Immediate Operational Win)

Time to Receive Welcome Letter (mean = 7.76) is the only parameter below 8.0 and has the highest variance (Std = 2.69) indicating severe inconsistency. While it has a weaker correlation with satisfaction ($r = 0.19$) it functions as a hygiene factor - its failure creates negative final impressions even after an otherwise good loan experience.

- Automate welcome letter dispatch within 48 hours of loan disbursement.
- Implement SLA tracking with escalation triggers for delays beyond 72 hours.
- Target: raise mean score to 8.2+ and reduce standard deviation to under 1.8 within one quarter.

3. North Zone Recovery Programme

North zone records the weakest satisfaction (7.60) and NPS (+25.5) with a detractor rate of 19.1%-nearly double the overall average. This is a significant risk requiring targeted intervention.

- Conduct 10-15 qualitative depth interviews with detractor customers in North zone.
- Assign a dedicated quality manager to the North zone for a 90-day intervention.
- Set performance targets: reduce detractor rate from 19.1% to below 10% within two quarters.

MEDIUM PRIORITY

4. Passive-to-Promoter Conversion Strategy

With 39.5% of customers classified as passives, there is a significant conversion opportunity. Passives churn at three times the rate of promoters and are easily poached by competitors with marginal offers.

- Introduce proactive post-disbursement check-in calls for 7 and 30 days.
- Launch a personalized loyalty and referral programme targeting passives.
- Target NPS of +50 by converting 15% of current passives to promoters.

5. Replicate South Zone Best Practices Nationwide

South zone delivers NPS +61.5 with only 6.2% detractors significantly outperforming other zones. Understanding what drives this excellence and scaling it is a low-cost, high-return strategy.

- Commission an internal benchmarking study to identify South zone operational and cultural differentiators.
- Create a 'South Zone Playbook' and roll out to all zones within two quarters.

6. Enhance Process Transparency

Loan charges, sanction terms, and documentation form a 'process transparency' cluster (all r approximately 0.55). While already scoring above 8.3, improving this further will compound satisfaction gains.

- Develop plain-language summary documents for loan charges and sanction terms.
- Introduce a digital tracking dashboard so customers can monitor loan status in real time.

MAINTAIN ~ SUSTAIN STRENGTHS

7. Protect Staff Excellence

Staff dimensions (politeness, availability, knowledge) score 8.44 - the highest of all parameters. Continue investing in hiring, onboarding, training and retention. This would be the most visible competitive advantage and must not be eroded.

8. Investigate the Satisfaction Gap

All individual parameters score higher than overall satisfaction (7.91). This suggests customers apply a 'global evaluation' beyond the sum of individual touchpoints. A follow-up qualitative study using laddering techniques could uncover hidden drivers not captured by the current questionnaire.

PRIORITY ACTION MATRIX

Recommendation	Priority	Expected Outcome
Optimize Loan Processing Experience	HIGH	Satisfaction +0.8 pts; NPS +8 to +10
Fix Welcome Letter Delivery	HIGH	Sub-8.0 parameter eliminated
North Zone Recovery Programme	HIGH	Detractor rate from 19.1% to <10%
Passive-to-Promoter Conversion	MEDIUM	NPS target +50; churn reduction
Replicate South Zone Best Practices	MEDIUM	Zone score convergence; NPS lift
Enhance Process Transparency	MEDIUM	Process cluster scores to 8.6+
Sustain Staff Quality	MAINTAIN	Protect #1 ranked dimension at 8.44+
Qualitative Follow-Up Study	LOW	Uncover hidden satisfaction drivers

9. CONCLUSION

Where We Stand

The loan service provider is in a strong overall position, which indicates a mean satisfaction of 7.91/10, an NPS of +39.5 (Great band), and 81% of customers in the Top-2-Box reflect a healthy and largely loyal customer base. The business is performing well, but there is a clear and actionable path to becoming exceptional.

What is Working

- Staff quality is the standout strength: politeness, availability, and knowledge all score 8.44/10, the highest of any dimension
- South zone is proof that world-class performance is achievable within the same organisation (NPS +61.5)
- Loan processing speed and documentation support meet customer expectations consistently across all zones
- 81% of customers rate their satisfaction at 8 or above - 4 in 5 customers are highly satisfied

What Needs Fixing

- Welcome letter delivery (7.76/10) is the only sub-8.0 parameter — a fixable operational gap creating negative last impressions
- North zone detractor rate (19.1%) is nearly double the overall average and demands immediate intervention
- 39.5% passive segment is satisfied today but vulnerable to competitors tomorrow

Three Immediate Priorities

Action	Timeline
Fix welcome letter dispatch: automate within 48 hours of disbursement	30 days
Launch North zone recovery programme which target detractor rate below 10%	90 days
Invest in end-to-end loan experience improvement with highest-return initiative	by 2026

Closing Thought

The gap between good and great is not structural; it is operational. The staff foundations are strong, South zone has already proven what exceptional looks like, and the data clearly points to where effort should go next. With focused action on the three priorities above, a realistic target of NPS +50 and mean satisfaction above 8.5 is achievable within the next two to three quarters.

Thank You